Positional Games

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An extremely far-reaching generalisation of the simple game of Tic-Tac-Toe takes the following form. We have an arbitrary finite set as the board of the game, and some of its subsets are designated as winning sets. Two players alternately play on an unoccupied cell of the board, and the first player to complete a winning set is the winner — otherwise, the game ends in a draw. This is called the strong positional game (or strong game in short).

In the weak version, also called the Maker-Breaker version, the second player's aim is not to occupy a winning set but just to prevent the first player from doing so. For both versions, the interest is in the general results about the games. One may also focus on the more natural multidimensional generalisation of Tic-Tac-Toe: the Hales-Jewett game. Very roughly speaking, a fair amount is known for Maker-Breakers while almost nothing is known for strong games.

The aim of the course is to provide a brief introduction to this combinatorial discipline to those with interest and basic knowledge in combinatorics. We will be covering some very elegant results in the field, as well as many of its standing challenges and open problems. Among others, these include the Strategy Stealing argument, the Pairing Strategies, the Erdös-Selfridge Theorem, the Pairing Conjecture and the Neighbourhood Conjecture.

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1 Introduction

We will set up the terminologies that will be used in future lectures.

Definition 1: Board & Winning Lines

We define the **board** X to be a finite set unless stated otherwise.
Let $\mathcal{F} \subseteq \mathcal{P}(X)$ be the family of **winning lines** or **winning sets**.

This family can be represented as a hypergraph on X, where elements of X are the vertices of the hypergraph while elements of $\mathcal{F}$ are the edges of the hypergraph.

Often, all $A \in \mathcal{F}$ have the same size n, i.e. $|A| = n$ for all $A \in \mathcal{F}$. In this case, $\mathcal{F}$ is a n-graph (a graph where all vertices have degree n).

For this course, we will cover two types of games:

Definition 2: Types of Games

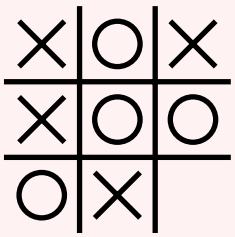
Strong Games
Two players takes turns to play until a player occupies all elements of some $A \in \mathcal{F}$. If X is filled with no winning line occupied by either player, then the game is considered as a draw.

Maker-breaker Games
Two players take turn to play. Player 1 wins if they occupy all elements of some $A \in \mathcal{F}$ while player 2 wins if player 1 is unable to do so.

Note that by definition, maker-breaker games cannot end in a draw. We say that a game is a P1 win if player 1 has a winning strategy and likewise for player 2.

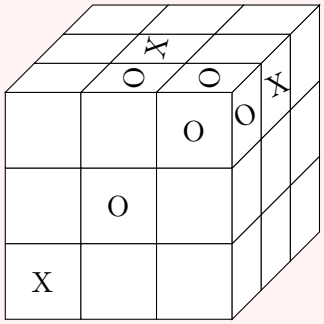
Example 1: Games

1. Normal tic-tac-toe



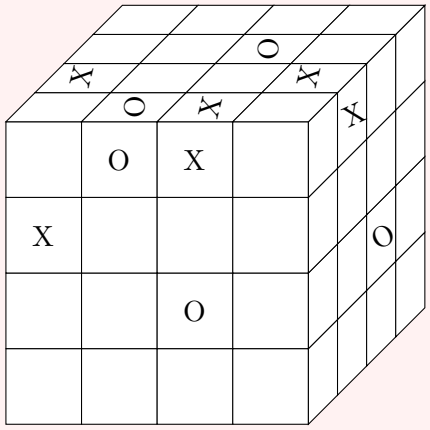
   This game is well-known to be a draw.

2. 3D tic-tac-toe
   - on a $3 \times 3 \times 3$ board



This game is known to be a P1 win.

- on a $4 \times 4 \times 4$ board



This version is still known to be a P1 win but the explicit winning strategy is very complicated.

Let's take a look at the generalisation of Tic-tac-toe in higher dimensions, that is the Hales-Jewett game of the $[n]^d-$game.

Definition 3: Hales-Jewett / $[n]^d-$game

Let the board $X = [n]^d = \{1, 2, \ldots, n\}^d = \{(a_1, a_2, \ldots, a_d) \mid a_1, a_2, \ldots, a_d \in [n]\}$.
The winning lines are the two type of lines defined as follows:

- A **combinatorial line** is a set of n points of the form:

$$\left\{ (x_1, x_2, \ldots, x_d) \;\middle|\; \begin{array}{l} x_i = x_j, \ \forall i, \ j \in I \\ x_i = a_i, \ \forall i \notin I \end{array} \right\}$$

  where $I \subseteq [d]$, $I \neq \emptyset$ and $a_i \in [n]$ for each $i \notin I$. Here, we refer I as the **active coordinates**. Simply speaking, combinatorial line is a "one-parameter set".

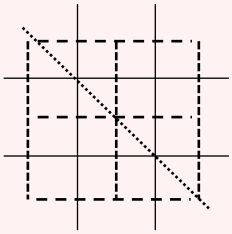
- A **line** is a set of the form:

$$\left\{ (x_1, x_2, \ldots, x_d) \;\middle|\; \begin{array}{l} x_i = x_j, \ \forall i, \ j \in I \\ x_i = x_j, \ \forall i, \ j \in J \\ x_i = a_i, \ \forall i \notin I \cup J \end{array} \right\}$$

  where $I, \ J \subseteq [d]$, $I \cup J \neq \emptyset$, $I \cap J = \emptyset$ and $a_i \in [n]$ for each $i \notin I \cup J$.

Example 2: Winning Lines on Tic-tac-toe

In normal tic-tac-toe ($[3]^2$-game), the combinatorial lines are the lines:



The main diagonal from the top left square to the bottom right square

$$\{(x, x) \mid 1 \leq x \leq 3\}$$

The 3 vertical lines

$$\{(1, x)\}, \{(2, x)\}, \{(3, x)\} \quad \text{where } 1 \leq x \leq 3$$

The 3 horizontal lines

$$\{(x, 1)\}, \{(x, 2)\}, \{(x, 3)\} \quad \text{where } 1 \leq x \leq 3$$

Meanwhile, the anti-diagonal line is not a combinatorial line (why?) and it is just a line with the form $\{(x, 3 - x) \mid 1 \leq x \leq 3\}$.

Example 3: More Lines

In $[4]^3$, both $\{(x, 1, 3) \mid 1 \leq x \leq 4\}$ and $\{(3, y, y) \mid 1 \leq y \leq 4\}$ are combinatorial lines.

In $[4]^4$, $\{(3, x, 1, 5 - x) \mid 1 \leq x \leq 4\}$ is a line.

This raises a question, how many winning lines are there in a $[n]^d$-game?

- When $d = 2$, we have $2n + 2$ winning lines. (n horizontal lines, n vertical lines and 2 diagonal lines)
- When $d = 3$, we have $3n^2 + 6n + 4$ winning lines. (left to the reader to figure out)

Theorem 1: Number of Winning Lines in $[n]^d$-game

In a $[n]^d$-game, there are exactly $\frac{(n+2)^d - n^d}{2}$ winning lines.

Proof

For $1 \leq j \leq d$, the j-th coordinate of the points in a winning line is either constant, increasing or decreasing (by definition of the winning lines).

So, for each coordinate, we have $n + 2$ options, which are $\{1, 2, \ldots, n, \text{Increasing}, \text{Decreasing}\}$. Thus, there are $(n + 2)^d$ possibilities. However, there are n^d sets where all coordinates are constants.

Furthermore, since each line is counted twice (once when increasing and once more when decreasing), we have $\frac{(n+2)^d - n^d}{2}$ winning lines.

We'll discuss two more games. Some graph theory knowledge is assumed.

Definition 4: Ramsey Graph Game

Starting on a complete graph on n vertices, K_n, each player take turns to colour the edges of K_n. The first player that manages to colour a monochromatic (1 colour) K_s, where $s \leq n$ wins.
For instance, if $s = 3$, the objective is to colour a monochromatic K_3 or a triangle.

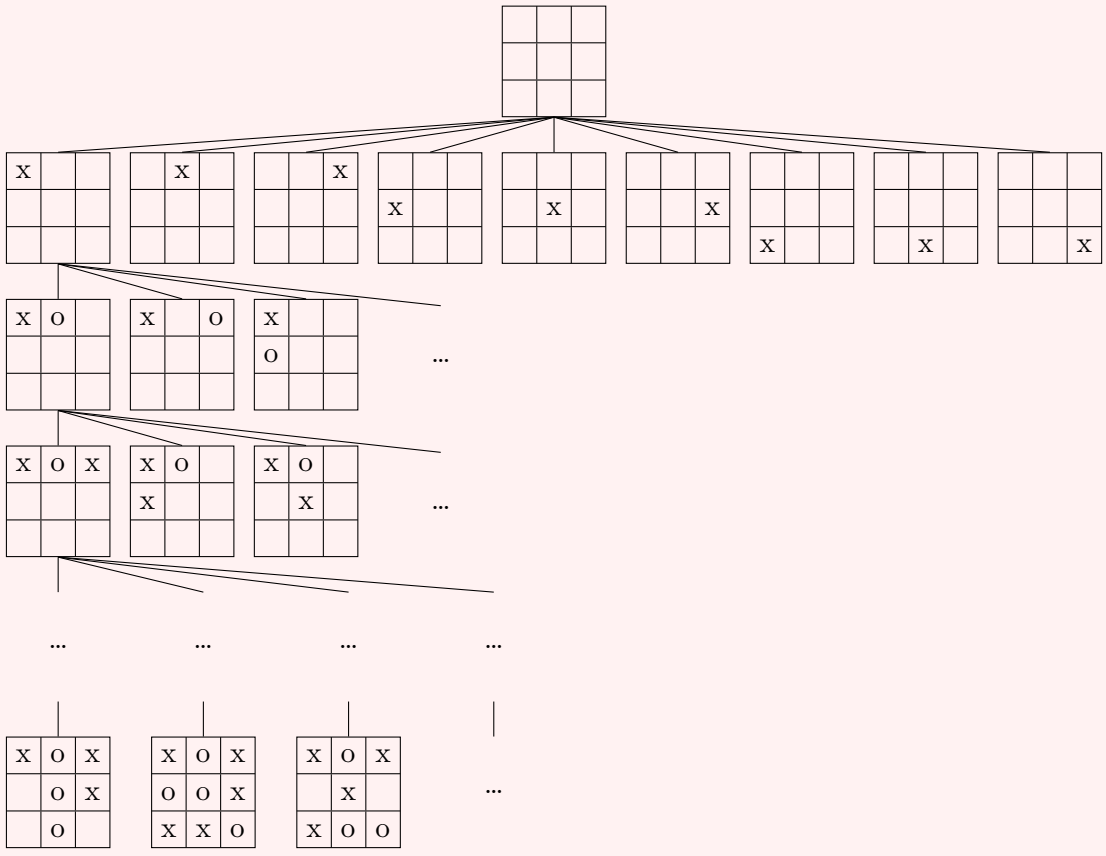
Definition 5: Binary Tree Game

Starting on a binary tree of n levels, the winning lines $\mathcal{F}$ are all paths from to initial root to the leaves.

What does it mean for a game to be winning for 1 player? In any game, every position has a value: one of "P1 win", "P2 win" or a draw, obtained from **backtracking** the leaves of **game tree** up until the position itself.

Example 4: Backtracking

In a game of tic-tac-toe with O being first, this position is a win for the player playing O. If we backtrack one move, any such position will still be a "O win". (Insert drawing Sri)



Let's formally define what it means for a player to have winning strategy:

Definition 6: Winning Strategy

A winning strategy for $P1$ is a function S : Positions $\rightarrow$ Move such that in any play of the game in which $P1$'s moves are determined by this function, $P1$ wins at the end.

In notation, a winning strategy for $P1$ means $\exists x_1 = S(\phi)$, $\forall y_1$, $\exists x_2 = S(x_1, y_1)$, $\forall y_2$, $\exists x_3 = S(x_1, y_1, x_2, y_2)$, ..., such that $P1$ wins.

We can define a winning strategy for $P2$ in the same way (replace $P1$ with $P2$).

Similarly, we can define a drawing strategy (no players can win) by replacing winning/wins with drawing/draws in the above definition.

It is important to note that a winning strategy for a player must work regardless of their opponent's moves.

In 1913, Ernst Zermelo [**?**] proved the following useful theorem in game theory:

Theorem 2: Zermelo's theorem

For every game, exactly one of the following hold:

- $P1$ has a winning strategy
- $P2$ has a winning strategy
- Each player has a drawing strategy

Proof

If $P1$ does not have a winning strategy, applying the negation to the logical definition of a winning strategy, $P2$ has a winning/drawing strategy.

Finally, let's look at a special type of games, infinite games.

Definition 7: Unrestricted n-in-a-row Tic-Tac-Toe

Let the board $X = \mathcal{Z}^2$ and $\mathcal{F} = n$ points in a row, either horizontally, vertically or diagonally.

If the entire sequence of moves results in no winning line for either player, the game is considered a draw. Note that a draw game last infinitely long.

In fact, in any such infinite game, with each winning line being finite (of size n), Zermelo's theorem still holds.

2 Strategy Stealing and Ramsey Criterion

Recall that a strong game is played on a hypergraph $(X, \mathcal{F})$ by two players, $P1$ and $P2$.

Theorem 3: Strategy Stealing

Any strong game cannot be a $P2$ win.

Proof

Suppose $P2$ has a winning strategy $\mathcal{S}$. Then $P1$ has a winning strategy (and thus provides the contradiction) as follows: $P1$ plays any move x_1 and pretends that x_1 is empty and he is the second player, playing with strategy $\mathcal{S}$. If $\mathcal{S}$ calls $P1$ to move to x_1, then he plays some other move x_2, and so on. Then $P1$ wins that play of the game as an extra square filled in cannot hurt him.

Remarks:

- The proof is completely non-constructive: it doesn't give a clue as to how $P1$ should play to guarantee at least a draw.
- Proof also applies when X is infinite.

Question: How can $P1$ win a strong game?

Sufficient condition: When a draw is impossible.

The necessary condition does not hold, eg. Binary tree.

Theorem 4: Ramsey Criterion

Suppose the game on $\mathcal{F}$ has no final drawing position (more generally, the hypergraph $\mathcal{F}$ has chromatic number at least 3). Then $P1$ has a winning strategy.

Note that strategy stealing applies to any kind of games in which the starting position is symmetry between P1 and P2 (meaning that both players want to occupy a winning set first), and an extra move doesn't hurt.

Example 5: Hex

Hex is a game invented by Piet Hein in 1942 and rediscovered by John Nash in 1948. The rules of the game are as follows: The game is played on a rhombus of hexagons of size $n \times n$, where $P1$ and $P2$ take the two opposite sides of the board each and mark alternately unmarked hexagons with their own colour/label. The player who marks a path connecting his/her opposite sides first wins the game. Usually the standard board size is 11×11.

It can be shown that a draw is impossible in a hex game. For precise statement and proof, see **Hex Theorem**.

Remark that the explicit first player winning strategy is not yet discovered for 11×11 Hex. In fact, no explicit winning strategy is found for $n \geq 8$.

Example 6: Bridg-It

The game was invented by Gale in 1960. The game is played on a square board consisting of a pair of interlaced $n \times (n+1)$ lattices (let say one with points $\circ$ and one with points $\times$). $P1$ connects adjacent $\circ$'s wanting a path from left to right, while $P2$ connects adjacent $\times$'s wanting a path from top to bottom. There is a restriction that no two edges are allowed to cross.

Note that Bridg-It is a variant of Hex game, and thus the game cannot end in a draw. However, an explicit first-player winning strategy for Bridg-It is known.

Let us look again Ramsey Graph Game introduced in Chapter 1. Ramsey Theorem says that any 2-coloured K_n contains a monochromatic K_s if n is large enough. The least such n is called **Ramsey number**, denoted by $R(s)$. Due to Erdős and Szekeres ([**?**, **?**]), we have the following bound for Ramsey numbers.

$$(\sqrt{2})^s \leq R(s) \leq 4^s.$$

Thus the game (K_n, K_s) is a $P1$ win if n is large enough (eg. $n \geq 4^s$).

Explicit Winning Strategy in Ramsey Graph Game:

Consider the game (K_n, K_3), where $n \geq 5$. Let x_1x_2 be the first edge coloured by $P1$. After the $P2$'s first move, there is a vertex $x \in \{x_1, x_2\}$ which is untouched by the edge coloured by $P2$. Since $n \geq 5$, there exists a vertex y untouched by both players. Let $u \in \{x_1, x_2\} \setminus \{x\}$. In the second move, $P1$ can colour uy, forcing $P2$ to colour xy. Then $P1$ can draw an uncoloured edge uv, where $v \notin \{x, y\}$. This gives two edges xv and yv, which can form a K_3 in $P1$'s next move. Therefore $P1$ wins (K_n, K_3).

We can see that an explicit winning strategy can be found in a complete graph with the number of vertices less than Ramsey number $R(3)$.

The case (K_n, K_4) is difficult to analyze, but a winning strategy is recently known by Bowler and Gut [**?**] in 2023. In fact, $P1$ has an explicit winning strategy on K_{17} (Note that $R(4) = 18$).

The case (K_n, K_s), $s \geq 5$ remains unsolved.

3 Winning and Drawing Strategies in Some Other Games

Why is finding a winning strategy so difficult? There are some troubles we are facing:

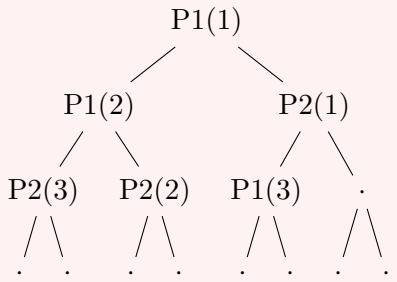
- $P1$ must create K_s to prevent $P2$ from creating an earlier K_s.
- A winning strategy for $P1$ corresponds to a statement of the form $\exists P1$ move: $\forall P2$ moves, $\exists P1$ moves, ... (The quantifiers go to a large depth).

Explicit Winning Strategies in Binary Tree Game:

It can be shown that $P1$ wins the game of binary tree of n levels, with time n. $P1$ starts by taking the root first, and then takes a child of the root, which $P2$ did not play in the subtree of this child. Continuing in this manner, $P1$ can obtain a path in n moves.

Example 7: Non-monotonicity of a $P1$ win Game

Non-monotonicity: Adding a winning line to a $P1$ win game may not result in another $P1$ win game. One of the examples is the binary tree of depth 4, plus the winning line (the three vertices labelled with $P2(1), P(2), P1(3)$ in the figure below). $P2$ can force a draw as follows: In the first move, $P1$ must start by taking the root (otherwise, $P2$ takes the root and $P1$ cannot form a winning line). Then $P2$ takes $P2(2)$, forcing $P1$ to take $P1(3)$ in the third move (otherwise $P2$ wins). Then $P2$ takes $P2(3)$, and the game is a draw.



$[n]^d$–Game:

Hales–Jewett Theorem says that any 2-coloured $[n]^d$ has a monochromatic line (even a combinatorial line, which is a more abstract object and does not depend on an ordering of $[n]^d$) if d is large enough. Let $HJ(n)$ denote the least d that works in the Hales–Jewett Theorem. So $[n]^d$-game is a $P1$ win if $d \geq HJ(n)$.

How large is $HJ(n)$? Obtaining a bound on $HJ(n)$ is rather difficult. Before describing the bound, we have to introduce the **Grzegorczyk hierarchy of primitive recursive functions**: Let $f_1(n) = 2n$. For $i \geq 2$, $f_i(n) = f_{i-1}(f_{i-1}(\cdots f_{i-1}(1)\cdots))$, this is the output where f_{i-1} is iterated n times. So $f_2(n) = 2^n$ and

$$f_3(n) = 2^{2^{\cdot^{\cdot^{\cdot^{2}}}}} \quad (n \text{ times})$$

which is called the **tower function**. Note that an equivalent statement of f_i is $f_i(n+1) = f_{i-1}(f_i(n))$ for $i \geq 2$. The best known upper bound, due to Shelah [**?**] in 1988, is $HJ(n) \leq f_4(n+4)$ which is enormous. We call f_4 the **super-tower function**. These f_k's are called the **"Grzegorcyk"** or **"Ackermann" hierarchy**.

Open question: Is $HJ(n) < f_3(n+1)$?

The best lower bound: $HJ(n) > \frac{cn^2}{\log n}$.

It is unlikely that the breaking points for $[n]^d$-game is anywhere close to $HJ(n)$, but not method is known for handling this game. Even the existence of a breaking point is questionable.

Explicit Winning Strategy in Some $[n]^d$–Game:

By case analysis, we can show that $[3]^2$–game and $[4]^2$–game are draws.

$[5]^2$–game is a draw, as follows:

12	1	8	1	11
6	2	2	9	10
3	7	#	9	3
6	7	4	4	10
11	5	8	5	12

In the labelling in $[5]^2$ board, there are two points on each winning line, disjointly (only one winning line has these two points). $P2$ can force a draw, by playing on the paired square to where $P1$ plays (note that if $P1$ plays #, $P2$ may just take any other square). Such a strategy is called a **pairing strategy**.

$[6]^2$–game is a draw by using a similar pairing strategy as follows:

13	9	5	6	9	14
11	2	2	#	3	12
8	#	#	#	3	8
7	4	#	#	#	7
11	4	#	1	1	12
14	10	5	6	10	13

For every $n \geq 5$, there exists a pairing strategy for $[n]^2$. It can be shown by passing n to $n+2$ as follows:

	bb	
a	$n \times n$	c
a	board	c
	dd	

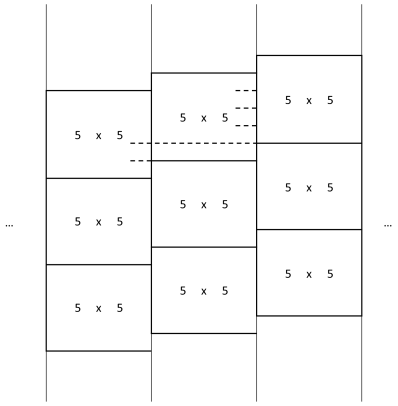
Some Studies on $[n]^d$-Games:

- $[3]^3$ is a P1 win (easy case analysis).
- $[4]^3$ is a P1 win [**?**].
- (Conjecture) $[5]^3$ is believed to be a draw.
- (Conjecture) $[5]^4$ is believed to be a P1 win.

Strategy in Some Unrestricted n-in-a-row Tic-Tac-Toe:

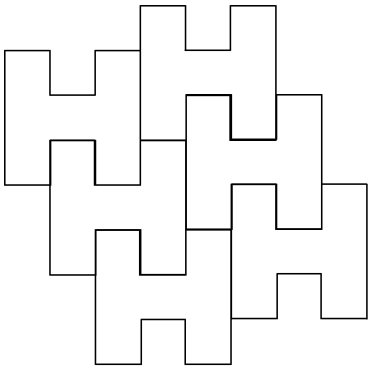
3-in-a-row and 4-in-a-row are easy P1 wins.

30-in-a-row is a draw. We tile $\mathbb{Z}^2$ with $[5]^2$ as follows:

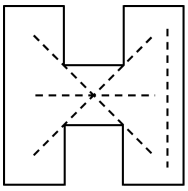


Any 30-in-a-row meets some $[5]^2$ in a winning line for the $[5]^2$-game. Then P2 can force a draw by playing his pairing strategy in any fixed $[5]^2$.

9-in-a-row is a draw. Tile $\mathbb{Z}^2$ with  as follows:



Any 9-in a rows meets some  in one of the following 4 lines.



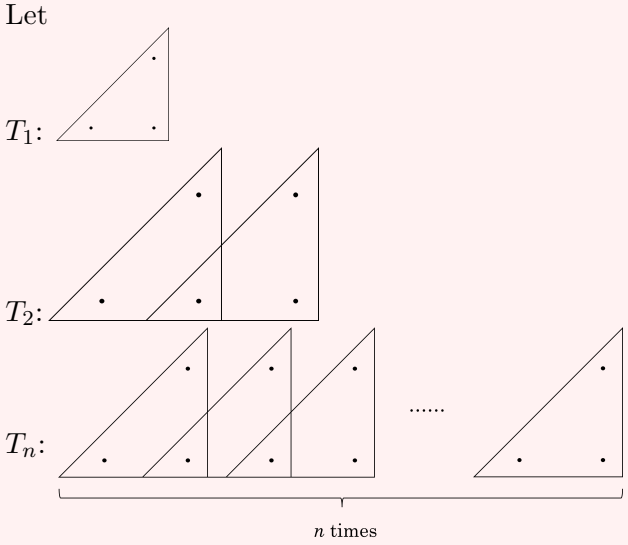
So P2 can stop P1 from filling one of these lines in any fixed  .

Remark: On an infinite board, we can have a game that is P1 win, but not in bounded time, i.e., there is no t such that P1 can guarantee to win in t moves. We can even have this when all winning lines have size $\leq n$, for some n, and the game has bounded maximum degree, i.e., each point is in $\leq k$ lines for some k.

Example 8: Binary Tree Game of Depth 4

Let B be the binary tree game of depth 4.

Let



We can allow our opponent to play in T_n twice before we need to respond. We might then need to keep responding n times, ending the game on T_n with a draw.
Example: Disjoint union of $B, T_1, T_2, \ldots$.

Hex:

No explicit winning strategy is known.

Bridg-It:

An explicit winning strategy was found by Oliver Gross. See his strategy in [**?**].

Unrestricted n-in-a-row:

- We do not know who can win in 7-in-a-row.
- (Conjecture) 6-in-a-row is believed to be a draw.
- (Conjecture) 5-in-a-row is believed to be a P1 win, but not in a bounded time.

If we play 5-in-a-row on a 15×15 board (Gomoku), then it is a P1 win.

4 Shannon Switching Game

Definition 8: Shannon Switching Game

Let G be a connected graph. P1 and P2 take turns to mark unclaimed edges of G. P2 wins if he marks all edges of a spanning tree, i.e., if his edges connects all of the vertices. Otherwise, P1 wins.

This game is also called **Connectivity Game**, where P1 and P2 are respectively called disconnector and connector.

For example, if G has a bridge, then P1 wins by taking the bridge in the first move.

Question: When does P2 win?
The answer was given by Lehman [**?**] in 1964.

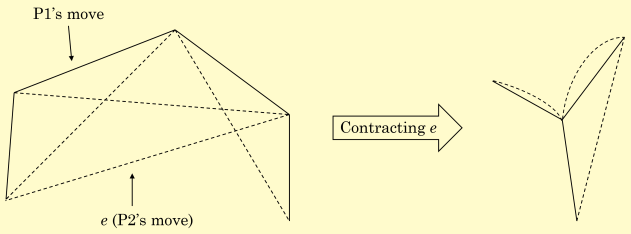
Proposition 1

P2 wins the Shannon Switching Game on a connected graph G if and only if G has two edge-disjoint spanning trees.

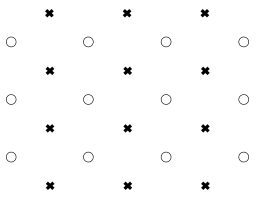
Proof

($\Rightarrow$) P1 plays any first move, then applies strategy stealing, i.e., ignores his first move, then pretends he is the second player, following P2's winning strategy. Then P1 makes a spanning tree, and it is disjoint from P2's spanning tree.

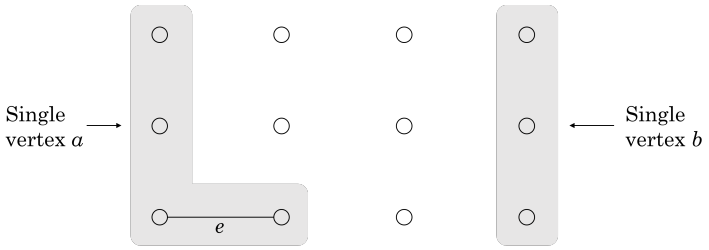
($\Leftarrow$) Whenever P1 takes an edge from one of the two spanning trees, thus splitting G into two components A and B. Then P2 chooses an edge e from the other spanning tree that joins A to B. Contracting e, we get a graph G' that still has two edge-disjoint spanning trees. So P2, by induction, can make a spanning tree in G'. Hence, with the presence of e, P2 makes a spanning tree in G.



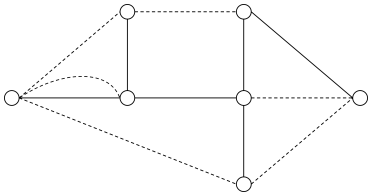
We can applying Lehman's proof to Bridg-It:



P1 wants to connect left to right. Each P2 move just deletes one allowed edge from the P1 graph. So we may simply play on the P1 vertices, with P2 claiming edges.



P1 plays the edge e. Since P1 wants to connect from a to b. Here we regard the four vertices highlighted on the left as a vertex a, and those on the right as a vertex b. In fact, P1 can connect all vertices in the (modified) graph by Proposition 4, because there exists two edge-disjoint spanning trees (represented by solid and dashed lines below).



Therefore P1 wins.

5 Disjoint Union of Games

Let $\mathcal{G}$ and $\mathcal{F}$ be the disjoint ground sets.

If $\mathcal{G}$ and $\mathcal{F}$ are both draw, then $\mathcal{G} \sqcup \mathcal{F}$ is a draw as follows: $P2$ just follows $P1$ into $\mathcal{G}$ or $\mathcal{F}$, respectively, playing his draw moves.

If $\mathcal{G}$ is a draw and $\mathcal{F}$ is a $P1$ win, then $\mathcal{G} \sqcup \mathcal{F}$ is $P1$ win as follows: $P1$ plays out his win in $\mathcal{F}$ and follows $P2$ into $\mathcal{G}$ whenever $P2$ plays there.

If $\mathcal{G}$ and $\mathcal{F}$ are both $P1$ win, then must be $\mathcal{G} \sqcup \mathcal{F}$ a $P1$ win?

Proposition 2

If $\mathcal{G}$ and $\mathcal{F}$ are both P1 wins, then so does $\mathcal{G} \sqcup \mathcal{F}$.

Definition 9

(i) For a hypergraph G, the **game-with-pass** on G, is the usual game on G, except that P2 can pass whenever he likes.

(ii) For a P1 win game G, the **delay** of G is the least k such that P1 can guarantee to win the game-with-pass on G, with P2 passing at most k times.

Proof

Suppose $\mathcal{G}$ has delay a, while $\mathcal{F}$ has delay b.

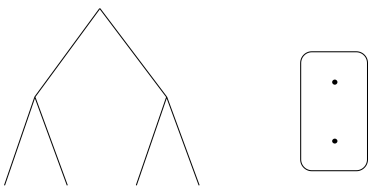
Let $\mathcal{S}$ be a strategy for the first player that guarantees a win with the other player passing at most a times.

Let $\mathcal{T}$ be a strategy for the second player that guarantees he can pass at least b times before losing.

Now P1 play as follows. P1 plays on $\mathcal{G}$ using the strategy $\mathcal{S}$, unless P2 plays in $\mathcal{F}$, in which case P1 replies on $\mathcal{F}$ using the strategy $\mathcal{T}$. If the strategy $\mathcal{T}$ calls on him to pass in $\mathcal{F}$, then he plays in $\mathcal{G}$ instead (according to $\mathcal{S}$, pretending P2 has just passed there).

If P2 wins this play of the game on $\mathcal{G} \sqcup \mathcal{F}$, then P1 passed at least b times in $\mathcal{F}$. So P2 passed at least b times in $\mathcal{G}$, and P1 has not yet won the game in $\mathcal{G}$. So P2 could now pass in $\mathcal{G}$, giving at least $b+1$ passes in $\mathcal{G}$, contradicting the definition of $\mathcal{S}$.

As a corollary, if $\mathcal{G}_1, \mathcal{G}_2, \ldots, \mathcal{G}_n$ are P1 win games, then $\mathcal{G}_1 \sqcup \mathcal{G}_2 \sqcup \cdots \mathcal{G}_n$ is P1 win. The result does not hold for infinite disjoint union. Let $\mathcal{G}$ be the disjoint union of the following two games:



Clearly $\mathcal{G}$ is P1 win. However $\bigsqcup^{\infty} \mathcal{G}$ is a draw, since P2 can keep playing one point on a pair of unclaimed points, forcing P1 to play on the pair.

6 Some Open Questions

(1) $[n]^d$ is a draw $\Rightarrow$ $[n+1]^d$ be a draw?

(2) $[n]^d$ is a P1 win $\Rightarrow$ $[n]^{d+1}$ is a P1 win?

(3) (K_N, K_s) is a P1 win $\Rightarrow$ (K_{N+1}, K_s) is a P1 win?

(4) It is known that (K_N, K_s) is a P1 win for all sufficiently large N (e.g. $N > R(s)$).What about infinite board, K_ω (this is an infinite complete graph, say, on $\mathbb{N}$). Is (K_ω, K_s) a P1 win?

(5) Is the infinite disjoint union of (K_N, K_s) games (e.g. $N \geq R(s)$) a P1 win?

(6) For a fixed s, can P1 win (K_N, K_s) in bounded time? Equivalently, is there a t such that for all $N \geq R(s)$, P1 wins (K_N, K_s) in at most t moves?

Some Progress:

- [?] Let $\mathcal{H}$ be a 5-uniform graph. Then $(K_\omega^{(5)}, \mathcal{H})$ is a draw.
- $\bigsqcup^\infty (K_N \sqcup K_n, \mathcal{G})$ is a draw.

Proposition 3

(4) is true if and only if (6) is true.

Note that in any play of the (K_N, K_s) or (K_ω, K_s), we may WLOG assume that the kth move is on $K_{[2k]}$.

Proof

(6) $\Rightarrow$ (4). Suppose we have a bound t, say. In particular, P1 can win (K_{2t}, K_s) within t moves (we can choose $t \geq \frac{1}{2}R(s)$ so that $2t \geq R(s)$). So let P1 follow a winning strategy for (K_{2t}, K_s) on K_ω board. Now we verify that this is well-defined: All the first t moves on K_ω are on $K_{[2t]}$. So P1 has won on K_ω in time t.

(4) $\Rightarrow$ (6). Suppose (6) is false. So for every t, there is a game $(K_{f(t)}, K_s)$ in which P2 can stay alive for time at least t, say following strategy $\mathcal{S}_t$.

Let P2 play as follows: P1 makes a move (has to be $\bullet^1 - \bullet^2$). P2 has two choices: join two new points, or join an old point to a new point. There are infinitely many of the $\mathcal{S}_t$ agree on what P2 should do, and P2 does that. Then P1 makes a move. Then P2 chooses a reply that infinitely many of "those infinitely many $\mathcal{S}_t$" agree on. (Again, there are infinitely many choices for P2). Continue.

At time T, P2 has followed infinitely many of the S_t, so in particular, has followed some S_t $(t > T)$. So P2 has lost by time t.

Remark: This argument is called **compactness argument**.

Question: (5) is true if and only if P1 has a winning strategy in $K_N \sqcup K_N$ such that P1 starts on a copy of K_N, keeps playing there, and can only play on the other copy of K_N after P2 plays there at least twice.

7 Maker-Breaker Game

Definition 10: Maker-Breaker Game

In the **maker-breaker** game on hypergraph $\mathcal{F}$, same rules as before except P1 (**maker**) wins if he occupies a line, and P2 (**breaker**) wins otherwise.

Example 9

In $[3]^2$:

2	2	3
	1	5
4		

Odd numbers represent P1's moves.
Even numbers represent P2's moves.

We do not case whether P2 occupies a line, so P1 wins the game.

Why study Maker-Breaker?

1. Cleaner version of strong game: Maker now only tries to make a line, not "make and block" at the same time.
2. Gives information about strong game. E.g. a breaker win $\Leftarrow$ a draw in the strong game.
3. Have monotonicity: An extra line cannot hurt maker.

   E.g., for all n, there exists a breakpoint d_0 such that $d \leq d_0 \Leftarrow [n]^d$ is a breaker win, and $d > d_0 \Leftarrow [n]^d$ is a maker win.

   E.g., for all s, there exists a breakpoint N_0 such that $N \leq N_0 \Leftarrow (K_N, K_s)$ is a breaker win, and $N > N_0 \Leftarrow (K_N, K_s)$ is a maker win.

Pairing Strategies:

Recall: a pairing strategy consists of a choice of two points in each line, distinctly. So the game is of course a breaker win.

The question is, when is there a pairing strategy?
Need: Any k lines contain at least $2k$ points.
This is also sufficient, by polygamous version of Hall's Theorem.

Hall's Theorem: to marry off some boys, each to a girl he likes, it is necessary and sufficient that any k boys like at least k girls (for all k).

To deduce polygamous Hall: marry off each boy at two girls he likes $\iff$ every k boys likes at least $2k$ girls.
Just replace each boy with two clones.

Remark: If each line has size n, then it is enough to have maximum degree at most $n/2$ (each point is in at most $n/2$ lines). Indeed, any k lines seeing ℓ points:

$$kn \leq \frac{\ell n}{2} \implies \ell \geq 2k \quad \text{(Double counting on points)}.$$

Proposition 4

There exists a pairing strategy in $[n]^d$-game if $d \le \frac{\log n - \log 2}{\log 3}$ (and so $d \le c \log n$).

Proof

The maximum degree is at most 3^d (each coordinate is increasing, decreasing, or constant). Based on the remark above, if there is a pairing strategy, then it must have $3^d \le n/2$. Equivalently,

$$d \le \frac{\log n - \log 2}{\log 3}.$$

How far could pairing work for $[n]^d$?
Need:

$$\begin{aligned} & 2 \times (\#\text{lines}) \le (\#\text{points}) \\ \iff & (n+2)^d - n^d \le n^d \\ \iff & (n+2)^d \le 2n^d \\ \iff & \left(1 + \frac{2}{n}\right)^d \le 2 \\ \iff & d \log \left(1 + \frac{2}{n}\right) \le \log 2. \end{aligned}$$

When n is large, $\log\left(1 + \frac{2}{n}\right) \approx \frac{2}{n}$. Thus there is no pairing strategy beyond $d = \frac{n \log 2}{2}$.

Remark: The bound 3^d for the maximum degree is overcounted. Actually, we have

$$\text{Maximum degree} = \begin{cases} \frac{3^d - 1}{2} & \text{if } n \text{ is odd,} \\ 2^d - 1 & \text{if } n \text{ is even.} \end{cases}$$

We only got $d = c \log n$ because the maximum degree in $[n]^d$ is much larger than the average degree.

We will reduce the maximum degree using **truncation**: throw out some points from each line (points where many coordinates equal). E.g.: $(1, 1, 1, n, 2, 2, 2, n-1, n-1, n-1, 5)$.

Proposition 5

In the $[n]^d$ game ($n \ge 12$), we can choose for each line L a subset L', with

$$|L'| \ge \frac{n}{2} - 1$$

such that each point belongs to at most $2d^{4d/n}$ of the various L' (The hypergraph formed by the lines L' has maximum degree $2d^{4d/n}$, and so the maximum degree of $[n]^d$-game is at most $2d^{4d/n}$).

Proof

Given a line L, then say that $k \in [n]$ is **overused** if the number of constant coordinates equal to k or $n + 1 - k$ is greater than $4d/n$.

The number of overused k is at most $n/2$ (if $> n/2$, then the number of constant coordinates is $> (n/4)(4d/n) = d$).

So we choose for each L, the subset

$$L' = \{x \in L : x \text{ is the } k\text{th point of } L,\ k \text{ is not overused, } k \neq (n+1)/2\}.$$

So $|L'| \geq n/2 - 1$.

Given $x \in [n]^d$, how many lines L are such that x is kth point of L and $x \in L'$?

Let

$$N_k := \{i : x_i = k \text{ or } n+1-k\}.$$

The active coordinates of line must be in N_k and actually are all but at most $4d/n$ of N_k. So

$$(\#\text{such lines}) \leq \left|N_k^{(\geq |N_k| - 4d/n)}\right| = \left|N_k^{(\leq 4d/n)}\right|.$$

So the degree of x is

$$\begin{aligned}
\sum_{k=1}^{n} \left|N_k^{(\leq 4d/n)}\right| &\leq \left|[d]^{(\leq 4d/n)}\right| \\
&= \sum_{i=0}^{4d/n} \binom{d}{i} \\
&\leq 2\binom{d}{4d/n} \quad (\text{as } 4d/n \leq d/3) \\
&< 2d^{4d/n}.
\end{aligned}$$

Corollary 1

In $[n]^d$ game for $n \geq 12$, there is a pairing strategy if $d \leq n/8$.

Proof

Find a pairing for the lines L' (instead of the lines L). Put $d = n/8$.

Note that the maximum degree is at most $2d4^{d/n}$, while each line contains at least $\frac{n}{2} - 1$ points.

So we are done (by the first remark) if

$$2d4^{d/n} \leq \frac{1}{2}\left(\frac{n}{2} - 1\right) \iff d^{4d/n} \leq \frac{n}{8} - \frac{1}{4}.$$

For $d = n/8$, we have

$$\sqrt{\frac{n}{8}} \leq \frac{n}{8} - \frac{1}{4},$$

which is correct if $n \geq 12$.

Conjectures on Maker-Breaker Game:

Hales-Jewett Conjecture: There is a pairing strategy in $[n]^d$-game if (#lines) $\leq \frac{1}{2}$(#points) (Unknown, but see later).

Gammill's Conjecture: Maker wins if and only if (#lines) $\geq$ #points (roughly $d \leq \frac{\log 3}{2} n$).

Citrenbaum's Conjecture: Maker wins if and only if $d \geq n$.

Ratio Conjecture: Hales-Jewett Conjecture said: if #lines $\leq \frac{1}{2}$#points (in $[n]^d$) then for all $S \subset [n]^d$ we have #lines in $S \leq \frac{1}{2}|S|$.

Equivalently, average degree $\leq n/2 \implies$ average degree in $S \leq n/2$.

Equivalently,

$$\frac{\#\text{lines}}{\#\text{points}} \leq \frac{1}{2} \implies \frac{\#\text{lines in } S}{|S|} \leq \frac{1}{2}.$$

This is a special case of the **Ratio Conjecture** (Patashnik, 1980 [**?**]): For all n, d and for all subsets $S \subset [n]^d$, we have

$$\frac{\#\text{lines in } S}{|S|} \leq \frac{\#\text{lines in } [n]^d}{|[n]^d|}.$$

More generally, is it true that if $|S| = n^e$, then $\frac{\#\text{lines in } S}{|S|}$ is maximised (over $S \subset [n]^d$) when S is an e-dimensional cube?

This is false!

Christofides [**?**] found a set S of size 3^5 in $[3]^d$ that has more lines than $[3]^5$ has. In fact, he found that the ratio conjecture is false for every n.

Proof: Proof for $n = 3$

We want to find $S \subset [n]^d$ such that

$$\frac{\#\text{lines in } S}{|S|} > \frac{\#\text{lines in } [3]^d}{3^d}.$$

Now we make some modifications on counting lines:

(i) Allow each line to be counted two times, i.e., can be counted each way. E.g.

$$\{(1,2,3),(2,2,3),(3,2,3)\}$$
$$\{(3,2,3),(2,2,3),(1,2,3)\}$$

are considered as two different lines.

(ii) Allow constant lines. E.g. the following $(1,2,3),(1,2,3),(1,2,3)$ is a line.

Now we want to show

$$\frac{\#\text{“lines” in } S}{|S|} > \frac{\#\text{“lines” in } [3]^d}{3^d}.$$

We will remove from $[3]^d$ the set T defined as

$$T = \{x \in [3]^d : (\#1 \text{ or } 3 \text{ that appear in } x) \text{ is } k\}.$$

So $T = \binom{d}{k}2^k$ (here k is fixed).

How many lines meet T?
How many lines meet T with midpoint $x \in T$?
E.g.

$$x = 111133333\underbrace{22222}_{\text{active coordinates here}}.$$

There are $\binom{d}{k}2^k 3^{d-k}$ such lines (2^k means we choose 1 or 3 by k times, and 3^{d-k} means we choose inc, dec, or constant $d-k$ times).

How many lines meet T with midpoint $x \notin T$? (so the endpoints are in T.)
E.g.,

$$x = \overbrace{111333\,\underbrace{222}_{\text{active}}}^{k}\ \underbrace{22}_{\text{constant}}.$$

There are $\binom{d}{k}4^k - \binom{d}{k}2^k$ such lines (The first term is found by choosing 1, 3, inc, or dec k times, while the second term is to avoid double counting constant lines).

Recall that the number of lines in $[3]^d$ is 5^d $(1, 2, 3,$ inc, dec$)$. So we are done if

$$\frac{\binom{d}{k}2^k 3^{d-k} + \binom{d}{k}4^k - \binom{d}{k}2^k}{\binom{d}{k}2^k} < \left(\frac{5}{3}\right)^d.$$

I.e., we want

$$3^{d-k} + 2^k - 1 < \left(\frac{5}{3}\right)^d.$$

Pick $k = \alpha d$ (α is fixed and d is large). Then we have

$$3^{(1-\alpha)d} + 2^{\alpha d} - 1 < \left(\frac{5}{3}\right)^d.$$

So we are done if

$$3^{1-\alpha} < \frac{5}{3} \quad \text{and} \quad 2^{\alpha} < \frac{5}{3},$$

i.e.,

$$\alpha > \frac{\log 9/5}{\log 3} \quad \text{and} \quad \alpha < \frac{\log 5/3}{\log 2} \quad (\text{such } \alpha \text{ exists}).$$

Pick α so that $\alpha d = k$ is integer. This completes the proof.

8 Solitaire Army

Definition 11: Solitaire Army

Let the board be an infinite plane $\mathbb{Z}^2$. Each point can have at most one man. Now the game plays as follows:

- In each move, a man jumps over one or more other men, horizontally or vertically (diagonally move is not allowed).
- The men jumped over are removed from the board.

Initial position: Mens are put at points (x, y) where $y \leq 0$. In other words, they are in the lower half plane.

Aim: Count how many men do we need to move a man forward to a point where $y \geq 1$.

- For $y = 1$, we need two men (put the pair in a vertical line).
- For $y = 2$, we need four men (put three men in a horizontal line, then put one behind the leftmost man just now).
- For $y = 3$, we need eight men (How?).
- For $y = 4$, we need twenty men (How?).
- For $y \geq 5$, it is impossible.

Now we show that it is impossible to move a man to a point in $y = 5$. The proof is from Conway's result from the early 1960s. We assign a **weight** w to each point with the following condition: if h_1, h_2 and h_3 are any consecutive points in a horizontal line or in a vertical line, then

$$w(h_1) + w(h_2) \geq w(h_3).$$

Clearly $w(h_1) + w(h_2) + \cdots + w(h_{k-2}) + w(h_{k-1}) \geq w(h_{k-2}) + w(h_{k-1}) \geq w(h_k)$.

Then we assign a **value** to each position P as follows: Take the sum of the weights of those points occupied by men in the position P.

When a men makes a jump, say from h_1 to h_k (means that the men on $h_2, \ldots, h_{k-1}$ are jumped over). In this move, the value of the new position can be found by replacing $w(h_1), \ldots, w(h_{k-1})$ by $w(h_k)$. The inequality above tells that the move cannot be an increase in the value of the new position. In other words, the target position cannot have a higher value.
Let w be a positive number satisfying $w + w^2 = 1$ (in fact, w is the golden ratio). Assume that we can move a man to a point where $y = 5$. Assign 1 to the point occupied by the last man in

the final position. Then we assign weights as below:

$$\begin{array}{ccccccccc}
 & & & & 1 & & & & \\
 & & & & w & & & & \\
 & & & & w^2 & & & & \\
 & & & & w^3 & & & & \\
 & & & & w^4 & & & & \\
\cdots & w^8 & w^7 & w^6 & w^5 & w^6 & w^7 & w^8 & \cdots \\
\cdots & w^9 & w^8 & w^7 & w^6 & w^7 & w^8 & w^9 & \cdots \\
\cdots & w^{10} & w^9 & w^8 & w^7 & w^8 & w^9 & w^{10} & \cdots \\
\iddots & & & & \vdots & & & & \ddots
\end{array}$$

The sum of all weights of points in the line $y = 0$ is

$$\begin{aligned}
w^5 + 2w^6 + 2w^7 + 2w^8 + \cdots &= w^5 + 2\sum_{k=6}^{\infty} w^k \\
&= w^5 + 2\frac{w^6}{1-w} \\
&= w^5 + 2\frac{w^6}{w^2} \\
&= w^5 + 2w^4 = w^3 + w^4 = w^2.
\end{aligned}$$

So the sum of all weights of points in the lower half plane is

$$w^2\left(1 + w + w^2 + w^3 + \cdots\right) = w^2\frac{1}{1-w} = w^2\frac{1}{w^2} = 1$$

which is the value of the final position. Hence no finite number of men will suffice to send a man to the line $y = 5$.

9 Erdős-Selfridge Theorem

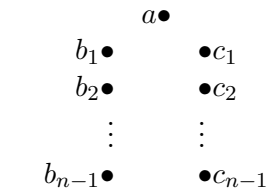
So far, the breakpoint for $[n]^d$-game: $cn \leq d_0 \leq$ iterated tower.

For (K_N, K_s)-game: $s+1 \leq N_0 \leq (4-\varepsilon)^s$.
To obtain the lower bound $s+1$, pairing strategy needs $\binom{N}{s} \leq \frac{1}{2}\binom{N}{2}$, this is false when $N \geq s+2$.

Question: Among all the Maker win game $\mathcal{F}$ all the lines of size n, what is the fewest lines in $\mathcal{F}$?
E.g. can achieve 2^{n-1} in both games:

(1) Binary tree game.

(2) The game



with winning lines $\{a, (b_1 \text{ or } c_1), (b_2 \text{ or } c_2), \dots\}$.

Can we get smaller than 2^{n-1}?

Theorem 5: Erdős-Selfridge Theorem

If $\mathcal{F}$ is an n-graph with $|\mathcal{F}| < 2^{n-1}$, then the Maker game on $\mathcal{F}$ is a Breaker win.

Remark:

(1) This is the best possible as there exists $\mathcal{F}$ with $|\mathcal{F}| = 2^{n-1}$ such that $\mathcal{F}$ is a Maker win.

(2) Given $|\mathcal{F}| < 2^{n-1}$, there exists a draw position: there exists an assignment of points to P1 or P2 such that neither has a winning line. Given a fixed winning line L, we get

$$\mathbb{P}(L \text{ is monochromatic}) = \frac{2}{2^n} = \frac{1}{2^{n-1}}.$$

So

$$\mathbb{P}(\text{there is a monochromatic line}) \leq \frac{|\mathcal{F}|}{2^{n-1}} < 1.$$

This means that we have a scenario that no line is monochromatic.

Quasiprobabilistic method: The proof can be summarised in the following ideas:

(1) To study the randomised version of the game (choice of "danger").

(2) Derandomisation by potential technique (the end of the proof does not involve random moves).

Randomised positional games: Let $|X| = N$ (vertex set). Then the number of plays is $N!$. Assuming that the board is filled in completely. Each player plays equally likely with probability $1/N!$. The outcomes are well-defined.

Heuristic argument:

- If overwhelming majority of the $N!$ plays ends with a Maker win, then we have a good chance that the game is a Maker win.
- Similarly for a Breaker win game.

Motivation: Let $|X| = N$. Then

$$\mathbb{E}(\#\text{monochromatic lines}) = \frac{\binom{N-n}{N/2}}{\binom{N}{N/2}} \cdot 2|\mathcal{F}| \leq 2^{-n} \cdot 2|\mathcal{F}| < 1.$$

Hence there is a play that ends with Breaker win.

In general, if $2^{-n+1}|\mathcal{F}| = c$, then there will be $\geq (1 - 1/c)N!$ of the $N!$ plays ends with a Breaker's win.

Proof

For a position P (Maker has played $x_1, \ldots, x_k$ and Breaker has played $y_1, \ldots, y_\ell$), we define the **danger** of P as

$$D(P) = \sum_{L \in \mathcal{F}} D(L),$$

where

$$D(L) = \begin{cases} 2^{-\#\text{empty in } L} & \text{if Breaker has not played in } L, \\ 0 & \text{otherwise.} \end{cases}$$

So $D(P)$ equals to the expectation of number of lines for Maker, if the board is filled in, each point at random. If the board is large, it is the same as saying "if subsequently players play randomly".

Let Breaker always move as to minimise the danger. Suppose it is Breaker's turn to move, in position P, with danger $D(P) = D$.

Claim: After a move each, danger $\leq D$.

Since the initial danger $< 1/2$, so after Maker's first move, the danger < 1 (as the worst case is that Maker's move is in every line, doubling the danger of each).

So after every Maker's move, the danger < 1, so Maker did not win.

Proof of Claim: Let say Breaker moves to y; Maker then moves to x.

Breaker's move: Then the danger decreases by

$$\sum_{L: y \in L} D(L) = \sum_{\substack{L: y \in L, \\ y \text{ is Breaker's} \\ \text{first move in } L}} 2^{-\#\text{empty in } L \text{ in position } P}.$$

Maker's move: Then the danger increases by

$$\sum_{L:x\in L} D(L) = \sum_{\substack{L:x\in L,\\ \text{Breaker has not played}\\ \text{in L at position P,}\\ \text{and } y\notin L}} 2^{-\#\text{empty in } L \text{ in position } P}.$$

But the danger decreased by Breaker $\geq$ the danger increased by Maker by the choice of y.

Threshold for Ramsey Game using Erdős-Selfridge Theorem: Let $\mathcal{F}$ be the winning lines in the game (K_N, K_s). So

$$|\mathcal{F}| = \binom{N}{s}, \quad n = \binom{s}{2}.$$

If $\binom{N}{s} < 2^{\binom{s}{2}} - 1$, then it is a Breaker win. By approximation, it is equivalent to say that

$$\frac{N}{s!} < 2^{\binom{s}{2}} - 1.$$

Using Stirling's formula, we obtain

$$N^s < \sqrt{2\pi s}\left(\frac{s}{e}\right)^s 2^{s(s-1)/2-1} \iff N < (2\pi s)^{\frac{1}{2s}}\left(\frac{s}{e}\right)^s 2^{(s-1)/2-1/s}.$$

Definition 12: Extremal game

We call a game on an n-graph $\mathcal{F}$ **extremal** if $|\mathcal{F}| = 2^{n-1}$ and it is Maker win.

Beck's Questions:

(1) Can we describe all extremal games?

(2) Is it true that Maker can win every extremal game in time n?

(2) is false: A. Sanders (2004) found extremal games that take exponential time ($\geq 2^{n-4}$ Maker's moves). Source: Trust me bro

Corollary 2

Breaker can win the (K_N, K_s)-game if $N \leq 2^{s/2}$.

Proof

We want

$$\binom{N}{s} < 2^{\binom{s}{2}-1}.$$

We have

$$\binom{N}{s} \sim \frac{N^s}{s!} \sim \frac{N^s}{s^s e^{-s}}.$$

So want

$$\frac{N}{se^{-1}} < 2^{\frac{s}{2}} \quad \text{i.e.} \quad N < 2^{\frac{s}{2}}\frac{s}{e}.$$

Thus the breakpoint N_0 has: $2^{s/2} < N_0 < (4-\varepsilon)^s$.

What about $[n]^d$-game?

We want

$$\frac{(n+2)^d - n^d}{2} < 2^{n-1}.$$

Since $n^{d-1} < (n+2)^d - n^d \leq (n+2)^d$, taking logarithm yields $d \log n < n \log 2$. Hence we have

$$d < \log 2 \frac{n}{\log n}$$

where this d is worse than pairing.

Definition 13: Simple game

We say that $\mathcal{F}$ is **simple** if any two lines meet in at most one point.

E.g. the $[n]^d$-game, but not the (K_N, K_s)-game.

Remark: For simple games, the upper bound of 2^n in Erdős-Selfridge Theorem can be replaced by (roughly) 4^n (more precisely, $4^{n-c\sqrt{n}}$).

Theorem 6: Maker form of Erdős-Selfridge

Let $\mathcal{F}$ be a simple n-graph on X ($|X| \geq 4$). If $|\mathcal{F}| > 2^{n-3}|X|$, then $\mathcal{F}$ is a Maker win game.

Proof

Consider the exact same danger function as in the proof of Erdős-Selfridge Theorem. Let Maker always move to maximise danger. Initially, danger $> |X|/8$.

Claim: After a Maker move, followed by a Breaker move, the danger decreases $\leq 1/4$.

Then we are done: There are $\leq |X|/2$ such moves, so final danger > 0, which means that the final position is a Maker win.

Proof of claim: Exactly the same as before except: danger decreases by $\leq 1/4$ on ≤ 1 line played by both Maker and Breaker.

If Maker plays x in L, and Breaker play y in L, then the danger changes from $\frac{1}{2^t}$ to $\frac{1}{2^{t-1}}$ (by Maker's move) and to 0 (by Breaker's move), for some $t \geq 2$.

Corollary 3

Maker wins $[n]^d$-game if $d \geq \frac{\log 2}{2} n^2$.

Proof

We want

$$\frac{(n+2)^d - n^d}{2} > 2^{n-3} n^d,$$

i.e.,

$$\left(1 + \frac{2}{n}\right)^d > 2^{n-2} + 1.$$

Hence we have

$$\frac{2d}{n} > (n-2)\log 2 \implies d > \frac{n(n-2)\log 2}{2}.$$

Thus the breakpoint for d_0: $\frac{n}{8} \leq d_0 < \frac{\log 2}{2} n^2$.

Remark: The general case for Theorem 9 is as follows:

Let $\mathcal{F}$ be an n-graph on X such that every two points of X are in at most d lines. If $|\mathcal{F}| < 2^{n-3} d|X|$, then $\mathcal{F}$ is a Maker win.

Now we will use Local Lemma to prove some results. Kindly refer to the book [**?**] for more details on Local Lemma.

Theorem 7: Lovász Local Lemma (Symmetric)

Let $A_1, A_2, \ldots, A_n$ be events such that $\mathbb{P}(A_i) \leq p$. Suppose for each event A_i, it is independent of all other events except for at most d of them. If $ep(d+1) < 1$, then

$$\mathbb{P}\left(\bigcap_{i=1}^{n} A_i^c\right) > 0.$$

Corollary 4

Consider a game on an n-graph. we can show that if each line meets $< 2^{n-3}$ lines, then there exists a draw position.

Proof

Assign points randomly.
For each line L, let A_L be the event that L is monochromatic. So

$$\mathbb{P}(A_L) = \frac{1}{2^{n-1}}.$$

Note that A_L and $A_{L'}$ are independent if and only if $L \cap L' = \emptyset$. Also, we have

$$e \cdot \frac{1}{2^{n-1}} \cdot 2^{n-3} = \frac{e}{4} < 1.$$

By Local Lemma, there is a draw position (no L is monochromatic).

Exercises:

(1) Let $\mathcal{F}$ be an n-regular n-graph. Show that there is a draw position.

(2) Let $\mathcal{F}$ be a 3-graph such that the maximum degree of $\mathcal{F}$ is at most $\frac{r^2}{3e}$ for some $r \in \mathbb{N}$. Show that $\mathcal{F}$ is r-colourable.

Neighbourhood Conjectures:

- Is there a $c > 0$ such that if each line meet $< c2^n$ other lines, then Breaker wins?
- Is there a $c > 0$ such that if maximum degree $< \frac{c2^n}{n}$, then Breaker wins?

Weaker question (game-theoretic local lemma question): is there a $c > 1$ such that if max degree $< c^n$, then Breaker win?

Turn it around: What is the smallest maximum degree in a Maker win game?

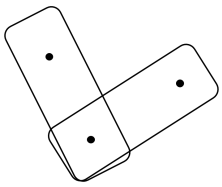
Known:

- $n/2$ (pairing draw): No
- 2^{n-1} (binary tree): Yes

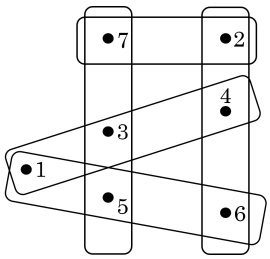
Nothing more is known for general n.

Studies on small n:

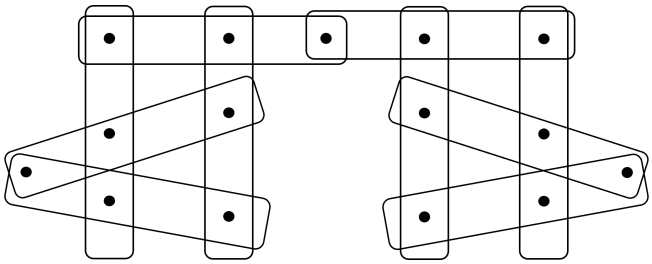
- $n = 2$.
  - Max degree 1: No
  - Max degree 2: Yes



- $n = 3$.
  - Max degree 1: No
  - Max degree 2: Yes
    Note that

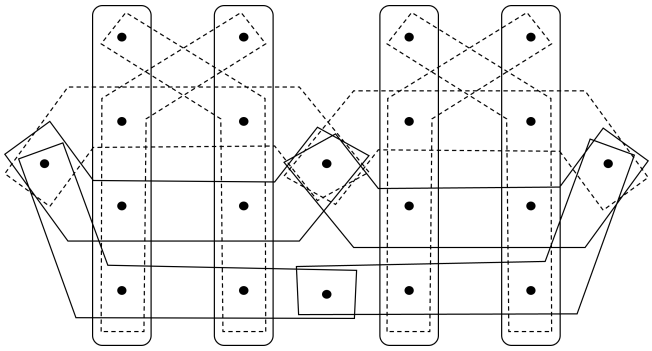


    is a Maker win (the numbers $1, 2, \ldots$ represent the turn of players and what points they pick), but some edges are not of size 3. So we can improve it by taking



- $n = 4$.
  - Max degree 2: No

– Max degree 4: Yes
This is done by doubling example for $n = 3$.



– Max degree 3: Yes (Done by [**?**])

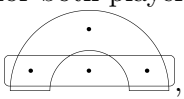
Theorem 8: (Beck)

Let $\mathcal{F}$ be a simple n-graph on X. If $|X| \leq 2^{n^2/100}$ and the maximum degree $\leq 1.5^n$, then Breaker wins. (Really, $\forall c < 2, \exists s > 0$ s.t. $|X| \leq 2^{sn^2}$ and max degree $\leq c^n$, then Breaker wins.)

We need:

(1) The **shutout game** (with parameter b) on a hypergraph $\mathcal{F}$.
Maker wins if he can occupy b points of some line $L \in \mathcal{F}$, when Breaker has occupied no points in L.

(2) The **shrinking shutout game** on $\mathcal{F}$: After a Maker move, he can declare some empty points as illegal for all later turns (for both players).

Note: This can hurt Breaker: e.g.  , $b = 2$. Maker can choose the leftmost point and make the rightmost point illegal. Then Maker must win because Breaker cannot prevent Maker from occupying two points in a line.

Lemma (Shutout form of Erdős-Selfridge): Let $\mathcal{F}$ be a hypergraph with $< 2^{b-1}$ lines. Then Breaker wins the b-shutout game on $\mathcal{F}$.

Proof

Define

$$D(P) = \sum_{L \in \mathcal{F}} D(L),$$

where

$$D(L) = \begin{cases} 0 & \text{if Breaker has played on } L, \\ 2^{-\#\text{empty}+|L|-b} & \text{otherwise.} \end{cases}$$

The initial danger is $< 1/2$. If Maker wins, then the danger is ≥ 1. Then we are done by proceeding exactly as in the proof of Erdős-Selfridge Theorem.

Lemma (Shrinking shutout form of Erdős-Selfridge): Let $\mathcal{F}$ be a hypergraph with $< 2^{b-1}$ lines, Then Breaker wins the shrinking shutout game on $\mathcal{F}$.

The proof is exactly the same as for the previous lemma.

We are ready to prove Beck's Theorem.

Proof

Fix an integer k (will be $n/10$).

Idea: Regard a line $L \in \mathcal{F}$ as **dangerous** if Maker has played $n-k$ points of this line, and Breaker has not played in it.

At all times, there will be a big board B and small board S, so

$$X = B \cup S.$$

After each Maker's move, some (unoccupied) points may move from B to S, i.e., B shrinks and S grows).

The Breaker will always play in B when the Maker does. Similarly, the Breaker will always play in S when the Maker does.

After each Maker move, if there exists a line $L \in \mathcal{F}$ such that Maker has played $n-k$ points in $L \cap B$ and no Breaker points in there, then move (up to) k empty points of L to S (some of these empty points might already be in S.)

The set of these k points is called a **dangerous** set.

On S: Breaker will play (if he can) a pairing strategy, by (hopefully) each dangerous set having two "private" points that are in no other dangerous sets.

On B: Breaker will play in such a way that neither of the following happen.

(A) Pairing fails. There exist dangerous sets $A, A_1, \ldots, A_{k-1}$ with

$$A = \{x_1, \ldots, x_{k-1}\}, \quad x_i \in A_i.$$

Say

$$A \subset L, \quad A_1 \subset L_1, \quad \cdots \quad A_{k-1} \subset L_{k-1}$$

where $L, L_1, \ldots, L_{k-1} \in \mathcal{F}$. So $L_1, L2, \ldots, L_{k-1}$ are **linked** (there is a line meeting them all), and on the big board B, Maker has occupied all but $k(k-1)$ points of $L_1 \cup \cdots \cup L_{k-1}$. But we have

$$|L_1 \cup \cdots \cup L_{k-1}| \geq n(k-1) - \binom{k-1}{2}$$

as $\mathcal{F}$ is simple.

So on the big board B, Maker has played $\geq (n-2k)(k-1)$ points of $L_1 \cup \cdots \cup L_{k-1}$.

(B) Maker has occupied the entire line $L \in \mathcal{F}$. Then $L \cap S$ has at least k points, or else there is a dangerous set from L at a earlier time.

This is exactly the same situation as (A) (a bit more).

So we are done by shrinking b-shutout Erdős-Selfridge on the hypergraph of all linked $\{L_1, \ldots, L_{k-1}\}$ with $b = (n-2k)(k-1)$, provided #linked sets $< 2^{b-1}$.

Let D be the maximum degree of $\mathcal{F}$. Note that

$$\begin{aligned}\#\text{linked sets} &\le |\mathcal{F}|\binom{n}{k-1}D^{k-1} \\ &\le |X|n^{k-1}D^k \quad \because |\mathcal{F}| \le |X|D.\end{aligned}$$

So done if

$$|X| < \frac{2^{(n-2k)(k-1)}-1}{n^{k-1}D^k}.$$

Some estimation (based on the hypothesis given):

$$\begin{aligned}\frac{2^{(n-2k)(k-1)}-1}{n^{k-1}D^k} &> \frac{2^{(n-2k)(k-1)}}{n^kD^k} \\ &= \frac{2^{(n-2k)k}}{n^kD^k2^{n-2k}} \\ &= \left(\frac{2^{n-2k}}{nD2^{n/k-2}}\right)^k \\ &= \left(\frac{2^{4n/5}}{n\cdot 1.5^n\cdot 2^{n/k-2}}\right)^{n/10} \qquad (\text{Set } k = n/10) \\ &> (2^{n/10})^{n/10} = 2^{n^2/100} \ge |X|.\end{aligned}$$

Corollary 5

The $[n]^d$-game is a Breaker win game if $d \le \frac{1}{5000}\frac{n^2}{\log n}$ (so Citrenbaum's Conjecture is false).

Proof

Apply Beck's Theorem to the set of truncated lines: we had each line has $\ge n/2 - 1$ points, and the max degree is $2d^{4d/n}$.

Note that $|X| = n^d$. Now we need $n^d \le 2^{(n^2/4)/100}$ and $2d^{4d/n} \le (1.5)^n$.

Taking logarithm gives

$$d\log n \le \frac{n^2}{400}\log 2 \quad \text{and} \quad \frac{4d}{n}\log d \le n\log 1.5.$$

Both are true if $d = \frac{n^2}{5000\log n}$.

Summary on $[n]^d$-games:

In strong game,

- if $d \leq \frac{n^2}{5000 \log n}$, then P2 has a drawing strategy;
- if $d >$ tower function of n, then P1 wins.

In Maker-Breaker game,

- if $d \leq \frac{n^2}{5000 \log n}$, then Breaker wins;
- if $d \geq \frac{\log 2}{2} n^2$, then Maker wins.